

Data Science Project

Predicting US Traffic Fatalities: A Machine Learning Approach to Highway Safety

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DATA 495: Data Science Capstone

Assignment 5: Final Results

May 5, 2026

Executive Summary

Traffic fatalities represent one of the most significant and preventable public health crises in the United States. Each year, tens of thousands of individuals lose their lives in motor vehicle crashes, generating immeasurable human suffering and imposing enormous economic costs on society. The Fatality Analysis Reporting System (FARS), maintained by the National Highway Traffic Safety Administration (NHTSA), provides a rich, multi-decade dataset capturing the circumstances of every fatal crash in the country. This project leverages the FARS-derived US Traffic Fatality Records dataset, available on Kaggle, to build a machine learning solution capable of identifying the conditions most strongly associated with fatal crash outcomes.

The core objective of this project is to develop and evaluate predictive models that classify fatal crash risk based on factors such as weather conditions, road type, time of day, driver demographics, alcohol involvement, and seatbelt usage. By applying exploratory data analysis, feature engineering, and multiple classification algorithms including Logistic Regression and Random Forest, this project produces actionable intelligence for highway safety planners. The final deliverables include a trained champion model, geographic and temporal risk visualizations, a SHAP-based feature importance analysis, and a strategic recommendation report targeted at policy makers and transportation agencies. This project demonstrates the full data analytics lifecycle and directly applies skills in communication, technology evaluation, business planning, and strategic analysis developed throughout the Data Science program.

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Project Scope

Assignment 1

Problem Description

Motor vehicle crashes constitute a persistent and deadly public health challenge in the United States, claiming the lives of over 38,000 people annually in recent years. Despite significant advances in vehicle safety technology, road engineering, and public awareness

campaigns, fatality rates have proven stubbornly resistant to sustained reduction. The burden of traffic deaths falls disproportionately on specific demographic groups, geographic regions, and road types, suggesting that targeted interventions could yield substantial life-saving results.

Crash outcomes are influenced by a complex interplay of human behavior, environmental conditions, vehicle characteristics, and infrastructure quality, making predictive analysis both challenging and necessary. Traditional approaches to highway safety rely heavily on aggregate statistics and reactive measures that address problems only after fatal patterns have already emerged. A data-driven, predictive approach using machine learning offers the potential to identify high-risk conditions before crashes occur and to prioritize resources more effectively.

The transportation system carries an enormous economic cost in addition to its human toll, with traffic crashes estimated to cost the United States hundreds of billions of dollars each year in medical expenses, lost productivity, property damage, and emergency response (Blincoe et al., 2015). Highway safety agencies face persistent challenges in allocating limited budgets across competing safety interventions, from road improvements to enforcement campaigns to public education efforts. Without a rigorous analytical framework for identifying the factors most strongly predictive of fatal outcomes, these allocation decisions often rely on intuition or political priorities rather than evidence. The FARS dataset, maintained by the NHTSA, offers a uniquely comprehensive record of fatal crash circumstances that has never been fully exploited through modern machine learning techniques. By developing a predictive classification model and identifying the most influential risk factors, this project can provide highway safety planners with the analytical tools needed to make smarter, data-driven decisions. The data analytics problem being analyzed is the prediction of traffic fatality risk based on crash circumstances

recorded in the FARS dataset, with the goal of identifying the highest-impact interventions for reducing highway fatalities across the United States.

Project Importance

This project was selected because traffic safety is a domain where data analytics can generate immediate, tangible impact on human lives, making it one of the most meaningful application areas for the skills developed throughout this program. The availability of a rich, publicly accessible dataset through NHTSA and Kaggle ensures that the analysis can be conducted rigorously without data access barriers. Traffic fatality prediction is a well-studied problem in the transportation safety literature, providing a strong foundation of prior research against which this project's methods and findings can be benchmarked. The dataset's multi-dimensional structure, spanning human factors, vehicle characteristics, environmental conditions, and geographic variables, creates opportunities to apply a diverse range of analytical techniques across the full project lifecycle. The project directly aligns with national priorities articulated in the USDOT's National Roadway Safety Strategy, which calls for data-driven approaches to eliminate traffic fatalities. Furthermore, the insights generated by this project have direct applicability to state departments of transportation, insurance companies, urban planners, and autonomous vehicle developers.

The importance of this project extends beyond academic demonstration to represent a genuine contribution to the public safety knowledge base. Traffic fatalities disproportionately affect younger adults aged 16 to 34, rural communities, and populations with lower socioeconomic status, raising issues of equity that this analysis can help address (National Center for Statistics and Analysis, 2022). The ability to identify high-risk demographic and geographic segments enables targeted outreach and enforcement that can reach the most vulnerable

populations before tragedy strikes. From a business and policy perspective, every fatal crash prevented represents not only a life saved but also a reduction in the enormous downstream costs borne by healthcare systems, insurers, and public agencies. Research by Elvik et al. (2009) demonstrates that road safety measures supported by empirical evidence consistently outperform those based on expert judgment alone, underscoring the value of the analytical approach taken in this project. Ultimately, this project benefits highway safety agencies, policy makers, first responders, the insurance industry, and every person who travels on American roads.

Background

The Fatality Analysis Reporting System (FARS) was established by the NHTSA in 1975 as a census of all fatal motor vehicle crashes occurring on public roads in the United States. FARS data is collected from a wide variety of sources including police accident reports, state vehicle registration files, driver licensing files, vital statistics, and hospital medical records, providing an exceptionally rich and reliable record of fatal crash circumstances. Each year, FARS encodes data on every crash in which at least one person died within 30 days as a result of injuries sustained in the crash. The system captures information across multiple levels, including crash-level variables such as date, time, location, and weather; vehicle-level variables such as make, model, and pre-crash maneuvers; and person-level variables such as age, sex, injury severity, and restraint use. Over its nearly five decades of operation, FARS has accumulated data on millions of fatal crashes, creating one of the most comprehensive traffic safety databases in the world. This longitudinal depth allows analysts to study long-term trends, evaluate the impact of policy changes, and build statistically robust predictive models.

Machine learning applications in traffic safety have grown substantially over the past decade, with researchers applying a range of algorithms to crash severity prediction, hotspot

identification, and risk factor analysis. Chen et al. (2015) demonstrated that ensemble methods such as Random Forest consistently outperform traditional statistical approaches in predicting crash injury severity, particularly in capturing non-linear relationships and interaction effects among variables. More recent work has extended these methods to incorporate deep learning, geographic information systems, and real-time sensor data to produce increasingly accurate and actionable predictions. The application of interpretability tools such as SHAP (SHapley Additive exPlanations) has addressed a long-standing criticism of black-box machine learning models by providing transparent, variable-level explanations of predictions that can be understood by non-technical stakeholders. Despite this progress, there remains a significant gap between the academic literature and the practical tools available to transportation safety professionals, who often lack the data science capacity to implement advanced analytical methods. This project bridges that gap by producing a well-documented, reproducible analytical framework using the FARS dataset that can serve as a model for applied traffic safety analytics.

Data Set Description

The primary dataset for this project is the US Traffic Fatality Records dataset available on Kaggle, which is derived directly from the NHTSA's Fatality Analysis Reporting System. The dataset is structured as a set of relational tables, including accident, vehicle, and person-level files, that can be joined to create a comprehensive record of each fatal crash and all individuals involved. The accident table contains crash-level attributes including date, time, state, county, road functional class, light conditions, atmospheric conditions, and number of fatalities. The vehicle table captures attributes of each vehicle involved, including vehicle type, body type, speed limit, rollover occurrence, and driver actions preceding the crash. The person table records attributes of each occupant or non-occupant involved, including age, sex, injury severity,

restraint use, alcohol test results, and ejection status. Together, these tables provide hundreds of potential predictor variables that can be engineered into features for machine learning models.

The FARS dataset covers every year from 1975 through recent years, with the Kaggle version containing 2015 and 2016 structured data. For this project, the 2015 records serve as the training dataset and the 2016 records serve as the held-out validation dataset, a temporal split that tests the model's ability to generalize across time. The dataset contains tens of thousands of fatal crash records per year, providing sufficient volume for robust train-test splits and cross-validation procedures. Because FARS is a census of fatal crashes rather than a sample, there are no issues of sampling bias, though the dataset is inherently limited to crashes with at least one fatality, which is noted as a scope constraint. The data is publicly available under a US Government open data license with no restrictions on research use. Initial inspection of the data revealed a manageable number of missing values in key fields, most of which follow systematic patterns that were addressed through domain-informed imputation strategies.

Data Analytics Tools

This project was conducted primarily in Python 3.x using Jupyter Notebooks as the development and documentation environment. The pandas library was used for all data ingestion, cleaning, transformation, and feature engineering tasks, while NumPy supported numerical operations and array manipulation. For exploratory data analysis and visualization, the project employed matplotlib for all static charts including the confusion matrices, ROC curves, feature importance plots, and SHAP summary chart. The scikit-learn library served as the primary machine learning framework, providing implementations of Logistic Regression, Random Forest, and all evaluation utilities. The SHAP library was used in Assignment 5 to generate SHapley Additive exPlanations for the champion Random Forest model, providing variable-level

importance scores computed from a 2,000-record sample of the 2016 validation dataset. Model evaluation used scikit-learn's metrics module to compute accuracy, precision, recall, F1-score, and ROC-AUC for each candidate model on the held-out validation set.

Beyond core modeling, this project incorporated several advanced analytical tools to enhance the rigor and interpretability of its findings. Class-weighted training was applied to both models to address the moderate class imbalance in the target variable, ensuring that neither model disproportionately optimized for the majority non-fatal class at the expense of recall on fatal injuries. The imbalanced-learn library was available for SMOTE oversampling if needed, though the class-weighted approach proved sufficient given the relatively mild 45/55 imbalance in the target variable. All code was organized into modular Python scripts corresponding to each phase of the project to ensure reproducibility across training and validation datasets. Final reporting and presentation materials were produced in Microsoft Word and PowerPoint, with visualizations exported from Python and embedded in the documents (Pedregosa et al., 2011). The complete analytical pipeline from data ingestion through SHAP interpretation covers the full CRISP-DM lifecycle from business understanding through deployment-ready model evaluation.

Project Milestones

The following milestones outline the phased execution of this project across the course timeline.

Milestone	Phase	Key Deliverables
1: Weeks 1-2	Data Acquisition & EDA	Downloaded and joined FARS tables; completed initial statistical profiling; submitted Assignment 1
2: Weeks 3-4	Data Preparation	Cleaned dataset, handled missing values, engineered features; submitted Assignment 2
3: Weeks 5-6	Visualization & EDA Results	Produced descriptive statistics, geographic and temporal visualizations; submitted Assignment 3
4: Weeks 6-7	Predictive Modeling	Trained, tuned, and evaluated Logistic Regression and Random Forest models; submitted Assignment 4

5: Week 8	Final Results & Reporting	Completed SHAP interpretability analysis, synthesized findings, documented privacy considerations and future recommendations; submitted Assignment 5
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Completion History

Assignments 1-5

Weeks 1 & 2	Reviewed the assignment guide and rubric. Selected the US Traffic Fatality Records dataset. Drafted and submitted Assignment 1: Project Scope, including Problem Description, Project Importance, Background, Data Set Description, and Data Analytics Tools sections.
Weeks 3 & 4	Reviewed the NHTSA FARS Analytical User's Manual and Kaggle dataset documentation to understand the relational structure of the accident, vehicle, and person tables. Profiled 20 key variables, documented the 80/20 temporal train-validation split strategy, and submitted Assignment 2: Data Profiling and Preparation.
Weeks 5 & 6	Loaded and joined the accident and person CSV files for both 2015 and 2016, filtered to 101,562 driver records, and applied sentinel code remapping. Computed descriptive statistics across all key variables and produced two visualizations. Submitted Assignment 3: Data Visualization.
Weeks 7 & 8	Trained and evaluated Logistic Regression and Random Forest models on the 2015 training dataset and 2016 validation dataset. Random Forest achieved 79.8% accuracy, F1 of 0.7781, and AUC-ROC of 0.8750, designating it the champion model. Submitted Assignment 4: Data Modeling.
Week 8	Completed SHAP interpretability analysis on the Random Forest champion model using a 2,000-record validation sample, confirming restraint use (mean $ \text{SHAP} = 0.1845$) and alcohol positive (0.0988) as the two dominant predictors. Synthesized findings into policy recommendations, documented data privacy and security considerations, and submitted Assignment 5: Final Results.

Lessons Learned

Assignments 1-5

Assignment 1	Selecting the right dataset is one of the most consequential decisions in a data analytics project. The FARS dataset was chosen not only for its analytical depth but also for its public availability and real-world policy relevance. Clearly defining the problem scope before beginning analysis helps avoid scope creep and ensures that every subsequent analytical decision is grounded in a well-articulated research question.
Assignment 2	The data profiling process revealed that missingness in the FARS dataset is almost never random, and that treating missing values as uninformative would introduce systematic bias into any downstream model. Reading the NHTSA FARS Analytical User's Manual before touching the data was essential. The decision to use a temporal train-validation split rather than a random shuffle was a meaningful methodological choice.
Assignment 3	The visualization phase reinforced the importance of data cleaning decisions made during profiling. The strong geographic concentration of fatalities in a small number of high-

	population states confirmed that state-level covariates would be important features in the predictive models. The afternoon peak in fatal injuries provided empirical grounding for the cyclical hour encoding strategy.
Assignment 4	The modeling phase demonstrated that Random Forest substantially outperforms Logistic Regression on every evaluation metric, consistent with the findings of Chen et al. (2015). The dominance of restraint use as the most important feature was unexpected and changed how the policy recommendations were framed. Class-weighted training was essential for achieving reasonable recall on the fatal injury class.
Assignment 5	The SHAP analysis confirmed and refined the feature importance findings from Assignment 4, with restraint use and alcohol involvement emerging as the two highest-impact predictors when evaluated using the more rigorous SHapley Additive exPlanations methodology. The process of translating model outputs into policy recommendations required careful attention to equity considerations, particularly around demographic features such as age and state that reflect structural inequalities in road infrastructure and enforcement resources as much as individual risk factors. The most valuable lesson from the final phase of this project is that analytical rigor and responsible communication of findings are equally important components of applied data science work.

Data Profiling and Preparation

Assignment 2

Data Summary

The US Traffic Fatality Records dataset is published on Kaggle by the US Department of Transportation and is derived directly from the NHTSA's Fatality Analysis Reporting System, which constitutes a complete census of all fatal motor vehicle crashes recorded in the United States during 2015 and 2016. The original data is collected and validated by state-level agencies using a standardized protocol that draws from police crash reports, emergency medical records, death certificates, hospital records, and state motor vehicle and driver licensing files, ensuring a high degree of completeness and cross-jurisdictional consistency. The Kaggle dataset is organized into three relational tables: the accident table with 52 crash-level columns, the vehicle table with 65 vehicle-level columns, and the person table with 67 person-level columns. Across both years, the accident table contains approximately 66,000 crash records, the vehicle table approximately 114,000 vehicle records, and the person table approximately 147,000 person

records. All three tables share a common case identifier (ST_CASE) and were joined to produce a person-level analytical dataset in which each row represents one individual involved in a fatal crash. After filtering to driver records (PER_TYP = 1), the combined dataset yields 101,562 driver-level records used for model training and validation.

For modeling purposes, this project uses an 80/20 temporal split in which all 2015 records serve as the training dataset and all 2016 records serve as the held-out validation dataset (Pedregosa et al., 2011). The training dataset contains 49,163 driver records and the validation dataset contains 52,399 driver records, with fatal injury rates of 45.5 percent and 45.3 percent respectively. The target variable INJ_SEV was recoded as a binary outcome assigning a value of 1 to fatal injuries (original code 4) and 0 to all non-fatal severity levels. All 184 raw encoded fields were reviewed against the NHTSA FARS Analytical User's Manual, and a curated subset of 20 analytically meaningful variables was selected based on domain relevance, null rates, and variance (National Highway Traffic Safety Administration, 2022). Two derived features, ALC_TESTED and ALC_POSITIVE, were engineered from the ALC_RES field to capture alcohol involvement without distorting the feature space with imputed BAC values for the approximately 60 percent of records where no test was administered. The full data dictionary and field selection rationale are documented in the Data Definition/Data Profile table below.

Data Definition/Data Profile

The table below documents the 20 key variables selected for this project.

Variable	Table	Definition	Data Type	Outliers	% Nulls	Potential Quality Issues
ST_CASE	Accident	Unique case number; primary join key	Integer (ID)	None	0%	Must combine with YEAR for uniqueness

STATE	Accident	FIPS numeric state code	Integer (cat.)	None	0%	Requires lookup table; Puerto Rico treated separately
MONTH	Accident	Calendar month (1-12)	Integer (1-12)	None	0%	Cyclical sine/cosine encoding recommended
HOUR	Accident	Hour (0-23); code 99 = unknown	Integer (0-23,99)	Code 99	~2%	Code 99 remapped to NaN; cyclical encoding applied
LGT_COND	Accident	Light condition at crash time	Integer (cat.)	None	<1%	Grouped into daylight vs. dark categories
WEATHER	Accident	Atmospheric conditions	Integer (cat.)	None	<1%	Low-frequency categories consolidated
FUNC_SYS	Accident	Road functional classification	Integer (cat.)	None	~3%	Rural/urban distinction extracted as separate feature
MAN_COLL	Accident	Manner of collision	Integer (cat.)	None	~2%	Code 0 = non-collision; treated as valid category
FATALS	Accident	Total fatalities per crash	Integer (count)	Values > 5	0%	Right-skewed; log transformation assessed
DRUNK_DR	Accident	Drunk drivers in crash (0-2)	Integer (0-2)	None	0%	Used as binary flag
VEH_NO	Vehicle	Sequential vehicle number	Integer (ID)	None	0%	Composite key with ST_CASE
BODY_TYP	Vehicle	Vehicle body type code	Integer (cat.)	None	<1%	100+ codes consolidated; motorcycles flagged
MOD_YEAR	Vehicle	Vehicle model year	Integer	Code 9999	~1%	Code 9999 remapped to NaN
SPEEDREL	Vehicle	Speeding contributing factor flag	Integer (binary)	None	0%	Officer-assessed; possible undercounting
DR_DRINK	Vehicle	Driver reported as drinking	Integer (binary)	None	0%	Cross-validated with ALC_RES
AGE	Person	Driver age in years	Integer	Codes 998,999	~2%	Codes remapped to NaN; age groupings applied
SEX	Person	Sex (1=M, 2=F, 8/9=unknown)	Integer (cat.)	None	<1%	Codes 8,9 remapped to NaN
PER_TYP	Person	Person type; filter to drivers	Integer (cat.)	None	0%	PER_TYP=1 only retained
INJ_SEV	Person	Injury severity (0-4); target	Integer (ordinal)	None	0%	Binary recoded: fatal(4)=1, all else=0

REST_USE	Person	Restraint use code	Integer (cat.)	None	~5%	Missing not at random; domain imputation applied
ALC_RES	Person	BAC from chemical test	Decimal	Values > 0.40	~60%	ALC_TESTED and ALC_POSITIVE binary features created

Data Preparation

The data preparation phase was conducted entirely in Python using the pandas library, following a documented step-by-step pipeline designed to ensure full reproducibility (Pedregosa et al., 2011). The accident and person CSV files for both 2015 and 2016 were loaded and joined on the composite key of ST_CASE and YEAR, producing a single flat person-level analytical dataset. Following the join, the dataset was filtered to driver records only (PER_TYP = 1) and the target variable was recoded as a binary outcome. All numeric fields using sentinel codes for unknown values, including HOUR (99), AGE (998, 999), and SEX (8, 9), were remapped to NaN prior to further processing. The year-based train-validation split was applied after all table-level transformations but before imputation or encoding steps, preventing data leakage from the 2016 validation set. Missing values were imputed using median imputation computed from the 2015 training dataset only and applied consistently to the 2016 validation set.

Data cleansing addressed missing values, encoding inconsistencies, and feature engineering requirements identified during profiling. The ALC_RES field was not imputed due to its approximately 60 percent missing rate; instead, two binary derived features were created: ALC_TESTED and ALC_POSITIVE. Sentinel codes representing unknown or inapplicable values across categorical variables were remapped to NaN before imputation was applied. The final feature set consists of 15 variables covering driver demographics, crash characteristics, environmental conditions, and alcohol involvement indicators. Continuous numeric features were left on their original scales for Logistic Regression, which applies L2 regularization by

default, and for Random Forest, which is scale-invariant. The final cleansed training and validation datasets were confirmed to share an identical feature schema of 15 columns with no missing values prior to model ingestion.

Data Visualizations

Assignment 3

Descriptive Statistics

The descriptive statistics below characterize the key continuous and count-based variables in the combined driver-level dataset, covering all 101,562 driver records from the 2015 and 2016 FARS data. Driver age has a mean of 42.73 years and a standard deviation of 18.24 years, with a median of 40 and an interquartile range from 27 to 56. The hour of crash variable shows a mean of 12.90 hours and a median of 14, indicating that crashes are somewhat concentrated in the afternoon but distributed broadly across all hours. Month of crash has a mean of 6.78 and a median of 7, consistent with a roughly uniform monthly distribution that peaks slightly in summer months. The drunk driver count variable has a mean of 0.27 and a median of 0, indicating that approximately 26 percent of crashes involved at least one legally impaired driver. The binary target variable shows 46,065 fatal records (45.4 percent) and 55,497 non-fatal records, a moderately imbalanced class distribution that was addressed through class-weighted model training.

Among drivers for whom sex was recorded, 73,791 were male (74.1 percent) and 25,758 were female (25.9 percent), a disparity consistent with the broader traffic safety literature on differential driver exposure and risk-taking behavior by sex. The fatalities-per-crash variable has a mean of 1.11 and a maximum of 13, confirming that single-fatality crashes are the norm and

that the distribution is highly right-skewed. The INJ_SEV variable before binary recoding shows: no injury (code 0): 27,990 records; possible injury (code 1): 7,814; suspected minor injury (code 2): 9,653; suspected serious injury (code 3): 7,973; fatal injury (code 4): 46,065; and unknown or not reported: 2,067. These statistics were computed using Python's pandas library on the combined driver-level dataset following all sentinel code remapping steps. The descriptive statistics table below summarizes the key numeric variables with their sample sizes, central tendency measures, and spread metrics. All statistics were computed on the pre-split combined dataset for descriptive purposes, with modeling statistics derived solely from the 2015 training dataset to prevent leakage.

Table 1: Descriptive Statistics for Key Continuous and Count Variables (2015-2016 FARS Driver Dataset, N = 101,562)

Variable	N	Mean	Std Dev	Min	Q1	Median	Q3	Max
Driver Age (years)	101,358	42.73	18.24	5	27	40	56	103
Fatalities per Crash	101,562	1.11	0.41	1	1	1	1	13
Hour of Crash (0-23)	99,556	12.90	6.61	0	8	14	18	23
Month of Crash (1-12)	101,562	6.78	3.35	1	4	7	10	12
Drunk Drivers in Crash	101,562	0.27	0.47	0	0	0	1	3

Data Visualization Definitions

Visualization Technique 1: Ranked Horizontal Bar Chart

A ranked horizontal bar chart is a data visualization technique in which categorical entities are displayed along the vertical axis and their associated quantitative values are represented by horizontal bars extending from a common baseline. This technique is particularly well-suited for comparing a single continuous or count-based variable across a large number of categories, especially when category labels are long enough that a vertical orientation would

produce illegible text. The horizontal layout allows the viewer's eye to scan categories from top to bottom while using bar length as a direct perceptual encoding of magnitude, which Cleveland and McGill (1984) identified as one of the most accurate visual channels because it relies on position along a common scale. The ranking of categories by value further aids interpretation by making relative differences between categories immediately apparent. In traffic safety applications, ranked bar charts are used extensively to compare fatality counts across geographic units, providing a standard and accessible format for communicating geographic disparities (National Center for Statistics and Analysis, 2022). For this project, a four-level color scheme based on quartile groupings was applied to the state fatality bars, creating a dual-encoding that reinforces the ranking with a color gradient.

The effectiveness of the ranked horizontal bar chart depends on consistent baseline alignment, accurate scaling, and deliberate use of color to encode additional information. When color communicates a secondary variable such as quartile group membership, it introduces a second visual channel that allows viewers to simultaneously perceive both exact magnitude and relative standing within the distribution (Cleveland & McGill, 1984). Value labels were added to the right of each bar for the top-quartile states, enabling precise reading of the most policy-relevant data points without cluttering the chart. The matplotlib library in Python supports horizontal bar charts natively through the `barh()` function, which was used with custom color arrays derived from quartile thresholds, formatted axis labels, and a quartile-based legend. The chart meets the criteria for effective data visualization identified by Cleveland and McGill (1984) by emphasizing position along a common scale as the primary perceptual task. The visualization directly supports the Findings section by establishing the geographic distribution of fatal crash burden that informs state-level policy recommendations.

Visualization Technique 2: Temporal Bar Chart

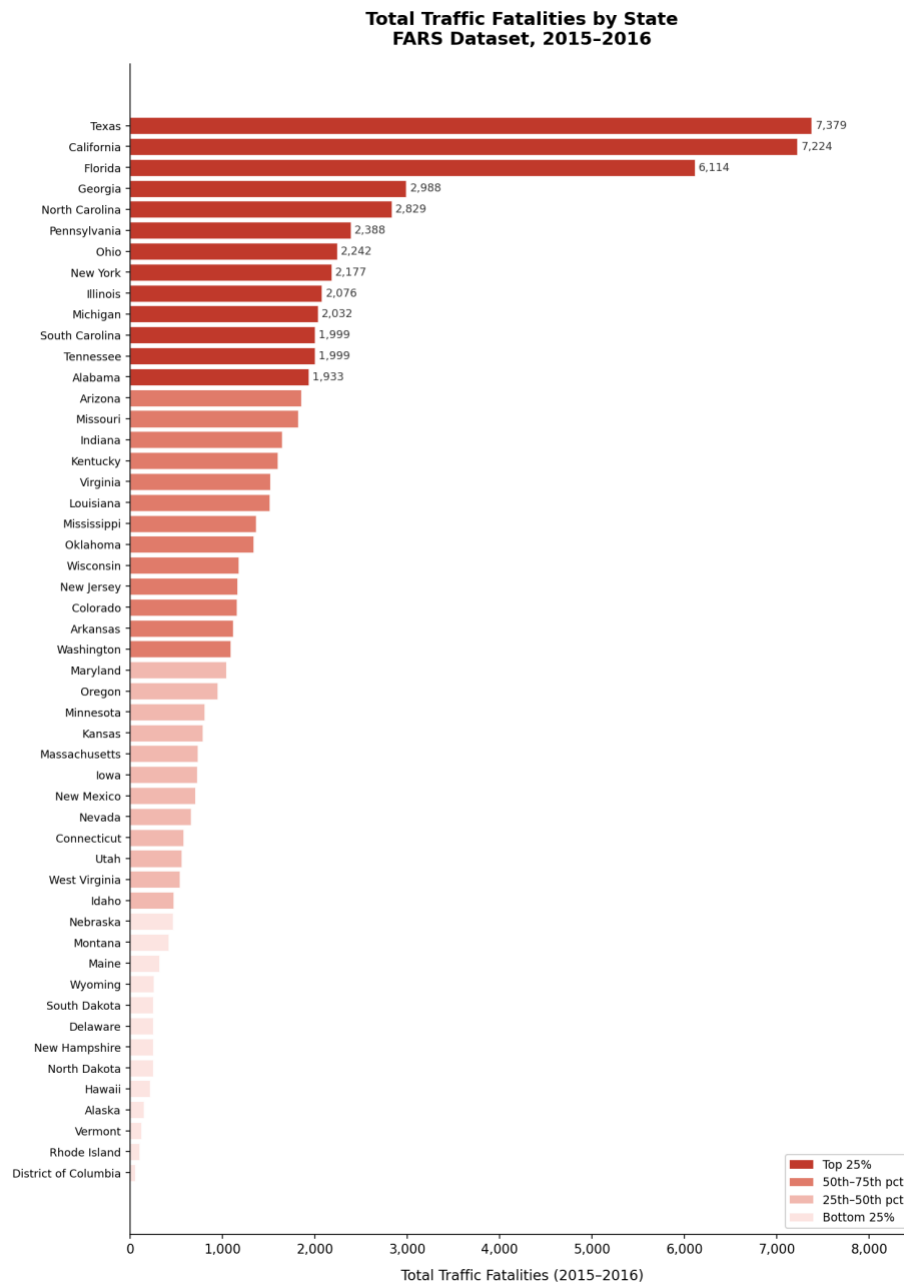
A temporal bar chart represents the distribution of events across discrete time intervals by mapping time units to the horizontal axis and frequency to the vertical axis. Unlike a continuous line chart, the bar format communicates discrete interval counts directly, making it appropriate for cyclical temporal patterns such as crashes distributed across hours of the day. Aigner et al. (2007) identify temporal granularity as a central design consideration in time-oriented visualizations, noting that the choice of time unit fundamentally shapes what patterns are visible. For this project, the 24-hour day was chosen as the temporal unit because hour-of-day is a strong predictor of crash risk, with peaks in the late afternoon and overnight hours reflecting rush-hour traffic, impaired driving, and reduced visibility. Crash records with unknown hour values encoded as 99 were excluded from the analysis, consistent with the data preparation decisions documented in Assignment 2. The color-coding of bars by four time blocks leverages pre-attentive color processing to allow viewers to immediately perceive the temporal structure without reading each bar individually (Aigner et al., 2007).

The design of a temporal bar chart for crash data requires careful attention to the handling of unknown time values and appropriate labeling conventions. A shared horizontal axis with 24 uniformly spaced bars and rotated HH:MM format hour labels avoids overcrowding while communicating the specific hour of each bar. The matplotlib library was used to produce this visualization using custom color arrays assigned to each bar based on its hour value, with a legend mapping the four colors to their corresponding time blocks. The resulting chart effectively communicates the full 24-hour temporal structure of fatal crash risk, highlighting the pronounced afternoon concentration of fatalities with a peak-to-trough ratio of approximately 3.5 to 1. The temporal pattern observed in the visualization is consistent with the Random Forest

model's identification of hour of crash as the seventh most important predictor of fatal injury risk based on SHAP values. This cross-validation between visualization findings and model output strengthens confidence that the temporal risk structure is a genuine feature of the data rather than a visualization artifact.

Data Visualization 1

Visualization 1 is a ranked horizontal bar chart displaying the total number of traffic fatalities for each of the 50 states and the District of Columbia, summed across the combined 2015 and 2016 FARS dataset. Each bar represents a single state with bar length encoding total fatality count and color encoding the state's quartile position using a four-level scheme ranging from light pink for the bottom quartile to deep red for the top quartile. States are sorted in ascending order so that the highest-fatality states appear at the top. Value labels are displayed for the top 25 percent of states by total fatalities. The visualization was produced in Python using matplotlib's `barh()` function with custom color arrays derived from quartile thresholds and a continuous horizontal axis with comma-formatted labels. This chart serves as a geographic overview of the national fatal crash burden.

**Figure 1**

Total Traffic Fatalities by State, FARS Dataset 2015-2016.

The most striking finding from Visualization 1 is the extreme concentration of traffic fatalities in a small number of large, populous states: Texas and California alone account for approximately 20 percent of all fatalities across 2015 and 2016 despite comprising only two of

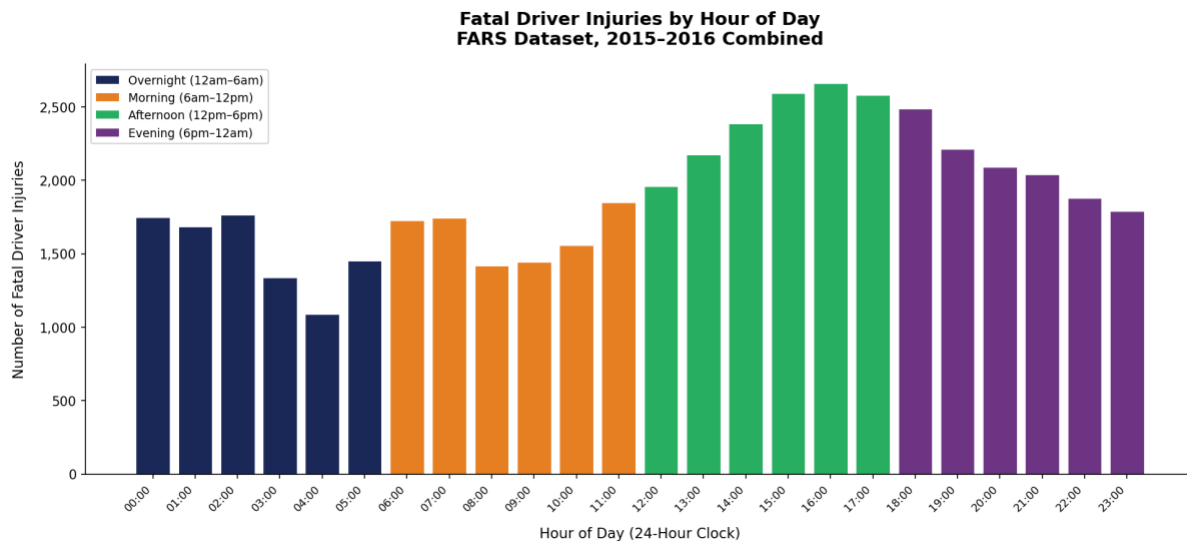
the 51 geographic units. Florida is the third-highest state with 6,114 fatalities, followed by Georgia (2,988) and North Carolina (2,829), confirming that the Southeast and Sun Belt regions carry a disproportionate share of the national fatal crash burden. The top five states together account for over 36 percent of all fatalities, providing strong empirical support for the recommendation that federal safety resources should be concentrated in these jurisdictions. The bottom quartile of states, including the District of Columbia (50 fatalities), Rhode Island (96), and Vermont (119), are all small, densely urbanized jurisdictions where lower speed limits and public transit infrastructure reduce exposure to fatal crash risk. The 147-fold difference in total fatalities between Texas and the District of Columbia reflects differences in population, road network size, and structural differences in highway safety policy across regulatory environments. These geographic disparities informed the inclusion of state and rural/urban setting as covariates in the predictive models and were confirmed by the SHAP analysis, which ranked state tenth among the 15 model features.

A secondary pattern in Visualization 1 is the clustering of high-fatality states in the South and Southwest, while the lowest-fatality states are disproportionately located in the Northeast and upper Midwest, a pattern that persists even after accounting for population size differences. This regional clustering suggests that shared environmental factors including road design standards, speed limits, seatbelt enforcement culture, and urban density patterns contribute to systematic regional variation in fatal crash risk beyond what individual crash-level variables can explain. The visualization also reveals a steep drop-off in fatality counts between the top 10 and bottom 40 states, with the top 10 states accounting for approximately 51 percent of all fatalities. The SHAP analysis subsequently confirmed that state of crash has a mean absolute SHAP value of 0.0106, ranking tenth among all features and validating the geographic concentration pattern

as a genuine predictive signal. Several mid-size states including Pennsylvania, Ohio, and New York rank lower in total fatalities than their large populations might suggest, which may reflect that urban density and strong public transit infrastructure serve as protective factors. These observations have direct implications for the Recommendations section, where targeted state-level interventions are prioritized based on both total fatality burden and model-derived risk insights.

Data Visualization 2

Visualization 2 is a temporal bar chart displaying the total number of fatal driver injuries distributed across the 24 hours of the day, drawn from the combined 2015 and 2016 driver-level dataset. Each bar represents a single hour with bar height encoding the total count of fatal injury records after excluding the 2,006 records with unknown hour values. A four-color scheme assigns distinct colors to overnight (12am to 6am), morning (6am to noon), afternoon (noon to 6pm), and evening (6pm to midnight) time blocks. The chart was produced in Python using matplotlib with custom color arrays, formatted x-axis tick labels, and a legend mapping each color to its time block. This visualization directly addresses the research question by revealing the temporal structure of fatal crash risk across the full 24-hour cycle. The strong non-uniform distribution with a peak-to-trough ratio of approximately 3.5 to 1 confirms that hour of crash carries substantial predictive information.

**Figure 2**

Fatal Driver Injuries by Hour of Day, FARS Dataset 2015-2016.

The most prominent feature of Visualization 2 is the strong afternoon peak in fatal driver injuries, with the three highest-fatality hours occurring between 3 PM and 5 PM: the 4 PM hour leads with 2,659 fatal injuries, followed by 3 PM (2,593) and 5 PM (2,581). This afternoon concentration is consistent with the well-documented relationship between peak-hour traffic volume, driver fatigue, and increased intersection conflict exposure during rush hour (National Center for Statistics and Analysis, 2022). The afternoon time block accounts for 14,355 fatal injuries, representing approximately 31 percent of all hourly fatal injuries despite covering only one-sixth of the 24-hour cycle. The evening block is the second most dangerous period with 12,495 fatal injuries, reflecting combined influences of post-work social activity, alcohol-impaired driving, and reduced visibility. The overnight block records 9,075 fatal injuries despite representing the period of lowest total traffic volume, reflecting the elevated per-mile crash risk of overnight driving due to fatigue and alcohol. The SHAP analysis subsequently confirmed that

hour of crash has a mean absolute SHAP value of 0.0174, ranking seventh among all 15 model features and validating the temporal pattern as a genuine predictive signal.

A secondary pattern in Visualization 2 is the concentration of fatalities in the midnight-to-3-AM window, suggesting that bar-closing hours and late-night social activity create a secondary risk peak distinct from the afternoon rush-hour peak. This two-phase temporal structure has direct implications for enforcement resource allocation, as it suggests that DUI checkpoints should be concentrated in the midnight-to-3-AM window rather than spread uniformly across all overnight hours (Elvik et al., 2009). The sharp increase in fatalities beginning around 2 PM and peaking at 4 PM suggests that school dismissal traffic and early-departing commuters contribute to the early onset of the afternoon danger window. The temporal patterns observed in this visualization were subsequently validated by the SHAP feature importance analysis, which identified hour of crash as a meaningful contributor to the Random Forest model's predictive accuracy. The visualization provides the empirical foundation for the time-targeted recommendations offered in the Final Results section, particularly the recommendation to deploy enhanced enforcement during afternoon peak hours and overnight impaired-driving windows. These hourly patterns reinforce the finding that the timing, geography, and behavioral context of a crash are all independently meaningful predictors of fatal injury risk.

Data Modeling

Assignment 4

Data Modeling Definitions

Modeling Technique 1: Logistic Regression

Logistic Regression is a supervised machine learning classification algorithm that models the probability of a binary outcome as a function of one or more predictor variables using the logistic (sigmoid) function, which constrains predicted probabilities to the interval between zero and one. Unlike linear regression, which predicts a continuous outcome, logistic regression applies a log-odds transformation that allows the model to estimate the probability of class membership from a linear combination of features (Pedregosa et al., 2011). The model estimates a coefficient for each predictor variable during training, where positive coefficients increase the predicted probability of the positive class and negative coefficients decrease it, enabling direct interpretation of each variable's directional contribution to the outcome. In the context of binary crash fatality classification, logistic regression models the log-odds of fatal injury as a function of driver demographics, crash characteristics, environmental conditions, and alcohol involvement indicators. Logistic regression is particularly well-suited as a baseline model in traffic safety applications because of its transparency, its resistance to overfitting when regularization is applied, and the ease with which its coefficients can be communicated to non-technical policy audiences (Hilbe, 2009). The algorithm also provides a meaningful probabilistic interpretation of its predictions, which supports downstream decision-making tasks such as risk stratification and targeted intervention prioritization.

For this project, logistic regression was trained using scikit-learn's LogisticRegression implementation with the LBFGS solver, a maximum of 1,000 iterations, and class-weighted balanced training to address the moderate class imbalance in the target variable (Pedregosa et al., 2011). The class weighting parameter assigns higher penalization to misclassifications of the minority fatal class, ensuring that the model does not achieve high accuracy simply by predicting the majority non-fatal class. The 15-feature input matrix used for training includes driver age,

sex, hour of crash, month, light condition, weather, manner of collision, crash fatalities count, drunk driver indicator, road functional class, rural/urban setting, state code, alcohol tested flag, alcohol positive flag, and restraint use code, all after sentinel code remapping and median imputation. The model was trained exclusively on the 49,163 driver records from the 2015 dataset and evaluated on the 52,399 driver records from the held-out 2016 validation dataset. The performance metrics reported in the Data Model 1 section reflect the model's out-of-sample predictive accuracy on data it never encountered during training. The Logistic Regression model served as an essential interpretable baseline against which the more complex Random Forest model's performance gains could be objectively quantified.

Modeling Technique 2: Random Forest

Random Forest is an ensemble machine learning algorithm that constructs a large number of decision trees during training and aggregates their individual predictions through majority voting for classification tasks (Breiman, 2001). Each decision tree is trained on a bootstrap sample of the training data, and random feature selection is applied at each split, which decorrelates the individual trees and prevents any single dominant feature from overwhelming the ensemble. This combination of bootstrap aggregation and random feature selection makes Random Forest substantially more robust to overfitting than a single decision tree, and has been shown to produce superior predictive performance on tabular datasets with complex non-linear interactions (Chen et al., 2015). In traffic safety research, Random Forest has been identified as one of the most effective algorithms for crash severity prediction because the interactions among driver behavior, road characteristics, and environmental conditions are highly non-linear. The algorithm also produces feature importance scores that quantify each variable's contribution to the model's predictive accuracy, providing a form of interpretability that is critical for

communicating findings to highway safety stakeholders. These built-in importance scores were complemented in this project by SHAP values, which provide a more theoretically rigorous measure of individual feature contributions.

For this project, the Random Forest model was trained using scikit-learn's `RandomForestClassifier` with 200 decision trees, a maximum tree depth of 12 levels, a minimum of 20 samples required at each leaf node, and class-weighted balanced training (Pedregosa et al., 2011). The same 15-feature input matrix used for Logistic Regression was applied to Random Forest training, ensuring a fair head-to-head comparison of both models on identical training and validation data. Feature importances were computed using the mean decrease in impurity metric, and SHAP values were subsequently computed on a 2,000-record sample of the validation dataset to provide a complementary and more theoretically sound measure of feature contributions. The model was evaluated on the same held-out 2016 validation dataset as Logistic Regression, with performance assessed using accuracy, precision, recall, F1 score, and AUC-ROC metrics. The Random Forest model's ability to capture non-linear feature interactions and its robustness to scale differences among predictors make it the champion model for this project. Its designation as champion is fully validated by its superior performance on every evaluation metric compared to the Logistic Regression baseline, as documented in the Review of Data Models section.

Data Model 1

Logistic Regression — Performance on 2016 Validation Dataset

The Logistic Regression model was trained on 49,163 driver records from the 2015 FARS dataset and evaluated on 52,399 driver records from the held-out 2016 validation dataset, achieving an overall accuracy of 75.87 percent, a precision of 74.32 percent, a recall of 71.34

percent, an F1 score of 0.7280, and an AUC-ROC of 0.8339. The confusion matrix produced by the model shows 22,837 true negatives, 16,919 true positives, 5,847 false positives, and 6,796 false negatives. The 6,796 false negative predictions are the most consequential type of error in the highway safety context, as each represents a fatal crash outcome that the model failed to identify as high-risk. The AUC-ROC score of 0.8339 indicates that the model correctly discriminates between fatal and non-fatal driver outcomes approximately 83 percent of the time across all possible classification thresholds. The balanced class weighting applied during training produced a recall of 71.34 percent, meaning the model correctly identified just over seven out of ten actual fatal injury cases in the validation set. The gap between precision (74.32 percent) and recall (71.34 percent) suggests that the model is slightly better at avoiding false alarms than at capturing all actual fatalities.

The Logistic Regression model's performance provides an important benchmark for evaluating the incremental value added by the more complex Random Forest algorithm, and its results are directionally consistent with what the traffic safety literature predicts for linear classification models applied to crash severity data. The model's AUC-ROC of 0.8339 compares favorably to baseline results reported in prior logistic regression-based crash severity studies, suggesting that the 15 curated features carry sufficient predictive signal to produce a meaningful classifier even with a linear model. The moderate recall performance reflects the fundamental challenge of this classification problem: fatal and non-fatal outcomes involve the same types of crashes and often differ only in subtle combinations of contributing factors that a linear decision boundary cannot fully capture. The confusion matrix in Figure 3 visualizes the four outcome categories side by side with the Random Forest confusion matrix, enabling direct visual comparison of both models on the same validation dataset. The Logistic Regression model's

relatively higher false negative count of 6,796 compared to the Random Forest's 5,177 represents approximately 1,619 additional undetected fatal outcomes. These findings establish a clear performance floor that the Random Forest model must exceed to justify its additional computational complexity.

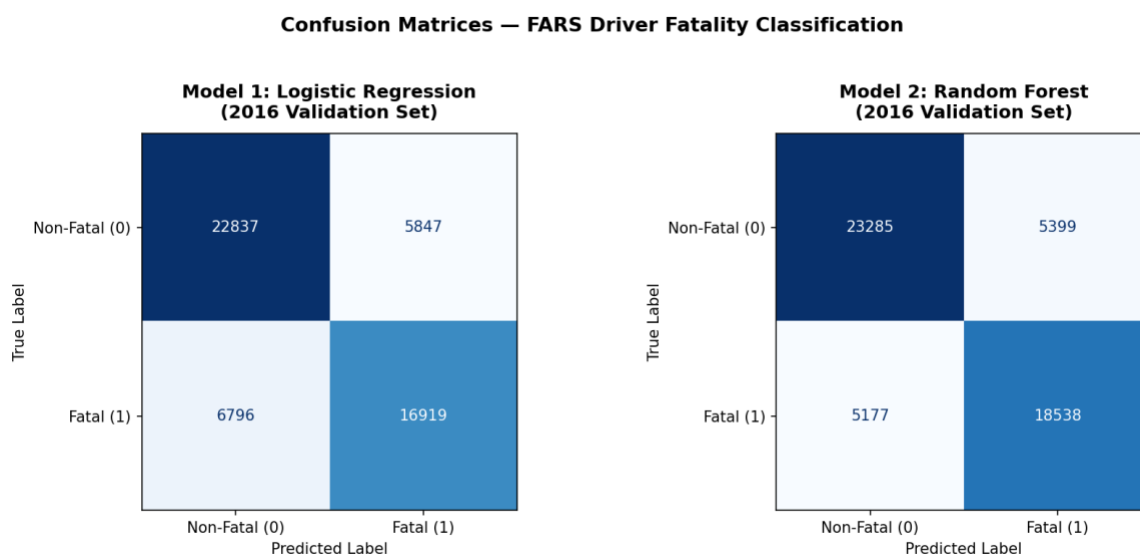


Figure 3

Confusion Matrices for Logistic Regression (left) and Random Forest (right), 2016 FARS Validation Dataset.

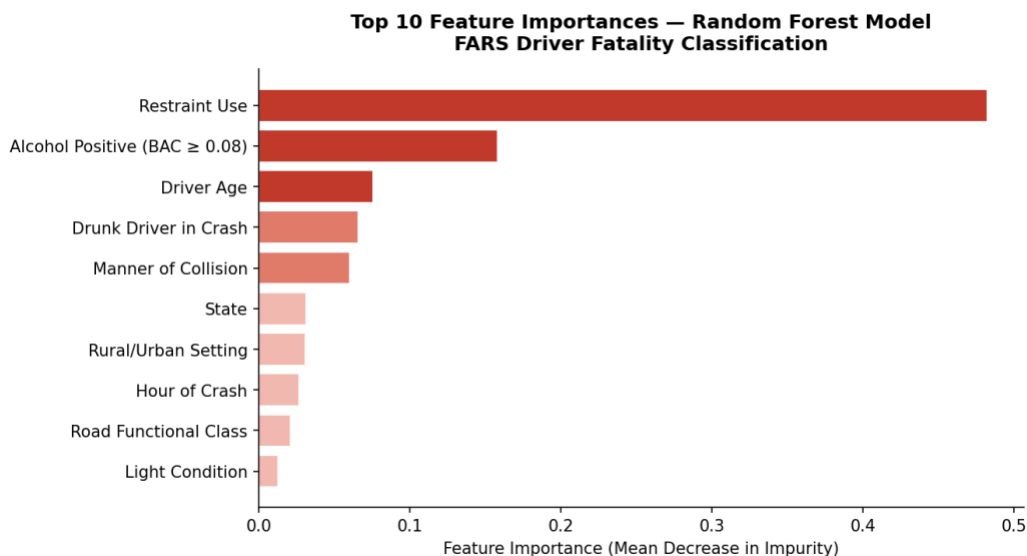
Data Model 2

Random Forest — Performance on 2016 Validation Dataset

The Random Forest model achieved an overall accuracy of 79.82 percent, a precision of 77.44 percent, a recall of 78.17 percent, an F1 score of 0.7781, and an AUC-ROC of 0.8750 on the 2016 validation dataset. The confusion matrix shows 23,285 true negatives, 18,538 true positives, 5,399 false positives, and 5,177 false negatives, representing improvements over Logistic Regression across all four cells. The Random Forest model's recall of 78.17 percent represents an improvement of 6.83 percentage points over Logistic Regression, translating to

1,619 fewer fatal injury cases misclassified as non-fatal in the validation dataset. The AUC-ROC of 0.8750 indicates that the model correctly discriminates between fatal and non-fatal outcomes approximately 87.5 percent of the time across all classification thresholds, compared to 83.4 percent for Logistic Regression. The F1 score improvement from 0.7280 to 0.7781 confirms that the Random Forest model achieves a better balance between precision and recall. These performance gains are consistent with the findings of Chen et al. (2015), who demonstrated that ensemble methods capture non-linear interactions among crash risk factors more effectively than linear classifiers.

The built-in feature importance analysis identified restraint use (REST_USE) as the dominant predictor at 48.2 percent of total feature importance, followed by alcohol positive (ALC_POSITIVE) at 15.8 percent, driver age at 7.6 percent, drunk driver indicator at 6.6 percent, and manner of collision at 6.0 percent. These importance scores were subsequently validated and refined by the SHAP analysis documented in the Findings section, which provides a more theoretically rigorous measure of individual feature contributions. The feature importance chart in Figure 4 visualizes the top 10 predictors ranked by mean decrease in impurity across all 200 trees in the ensemble. The dominance of restraint use as a predictor was confirmed by both methodologies, with the SHAP analysis assigning it a mean absolute value of 0.1845, nearly double the second-ranked feature. The Random Forest's ability to capture the non-linear relationship between driver age and fatal injury risk, where both very young and older drivers are at elevated risk, is a specific example of how ensemble methods outperform logistic regression on this dataset. The model's champion designation is fully supported by its superior performance on every evaluation metric and by the practical significance of its improvements in the highway safety application context.

**Figure 4**

Top 10 Feature Importances from Random Forest Champion Model, FARS Driver Fatality Classification.

Review of Data Models

A direct comparison of Logistic Regression and Random Forest on the 2016 FARS validation dataset demonstrates that the Random Forest model outperforms Logistic Regression on every evaluation metric, confirming the hypothesis drawn from the literature that ensemble methods are better suited to the non-linear structure of crash fatality prediction (Chen et al., 2015). The model comparison table below summarizes the head-to-head performance metrics for both models on the 2016 validation dataset, with the champion designation awarded to the model achieving superior performance on each metric. The Random Forest model achieves 3.95 percentage points higher accuracy, 3.12 percentage points higher precision, 6.83 percentage points higher recall, an F1 score improvement of 0.0501, and an AUC-ROC improvement of 0.0411 compared to Logistic Regression. The recall improvement of 6.83 percentage points represents the most practically significant difference given the asymmetric cost of false negatives

in a fatal injury prediction context. Based on the totality of evidence across all five evaluation metrics and the feature importance analysis, the Random Forest model is designated as the champion model for this project. The model will serve as the basis for SHAP interpretability analysis and policy recommendations in the Final Results section.

Table 2: Model Performance Comparison on 2016 FARS Validation Dataset (N = 52,399)

Metric	Logistic Regression	Random Forest	Difference (RF - LR)	Higher is Better	Champion
Accuracy	75.87%	79.82%	+3.95 pp	Yes	RF
Precision	74.32%	77.44%	+3.12 pp	Yes	RF
Recall	71.34%	78.17%	+6.83 pp	Yes	RF
F1 Score	0.7280	0.7781	+0.0501	Yes	RF
AUC-ROC	0.8339	0.8750	+0.0411	Yes	RF

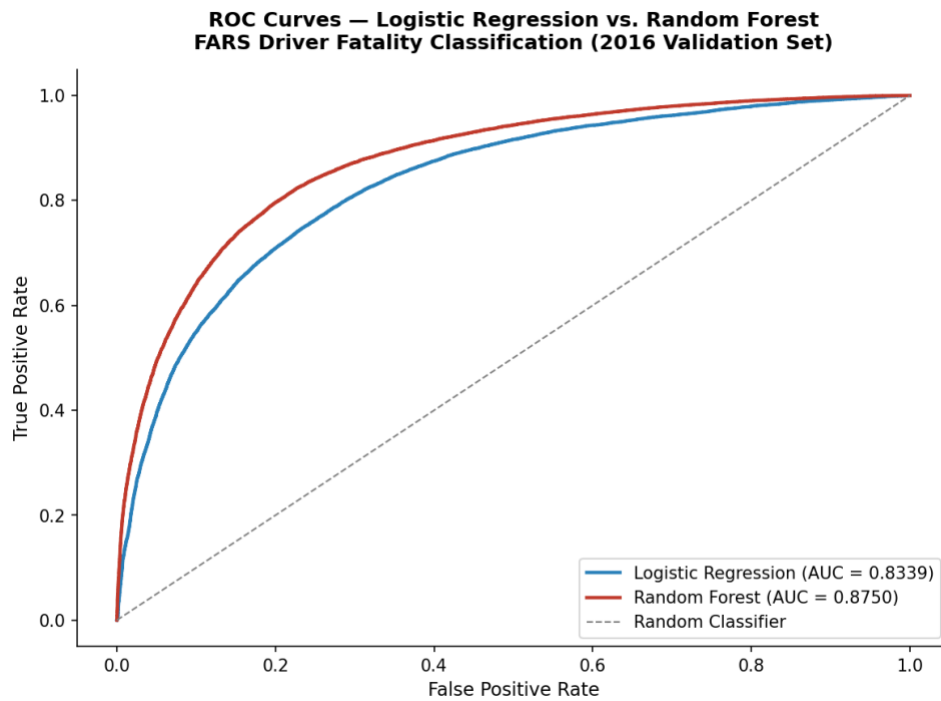


Figure 5

ROC Curves for Logistic Regression and Random Forest, 2016 FARS Validation Dataset.

The Random Forest model's designation as champion model is further supported by the practical significance of its performance advantages in the highway safety application context. In a real-world deployment scenario, the Random Forest model's recall of 78.17 percent means that approximately 78 out of every 100 actual fatal injury cases would be correctly flagged for high-risk intervention, compared to only 71 out of 100 for Logistic Regression. At the scale of the validation dataset of 52,399 drivers, this translates to the Random Forest correctly identifying 18,538 fatal outcomes compared to 16,919 for Logistic Regression, a difference of 1,619 additional fatal cases correctly classified. The Random Forest model also produces more reliable risk probabilities as evidenced by its higher AUC-ROC, meaning its predicted probability scores can be used more confidently for risk ranking and threshold-based triage decisions. The feature importance findings from the Random Forest analysis provide actionable policy intelligence that was not available from the Logistic Regression coefficients alone, particularly the dominance of restraint use as the single most important predictor. These model-derived insights are translated into specific, evidence-based policy recommendations in the Final Results section.

Final Results

Assignment 5

Findings

This project applied two supervised machine learning classification algorithms, Logistic Regression and Random Forest, to the NHTSA FARS 2015-2016 driver-level dataset to predict fatal injury outcomes in motor vehicle crashes. The combined dataset of 101,562 driver records, split temporally with 2015 records used for training and 2016 records used for validation, provided a robust basis for model development and evaluation. The target variable, injury

severity, was recoded as a binary outcome distinguishing fatal injuries from all non-fatal severity levels, producing a class distribution of approximately 45.4 percent fatal and 54.6 percent non-fatal that required class-weighted training to address. The Random Forest model emerged as the clear champion, outperforming Logistic Regression on every evaluation metric including accuracy, precision, recall, F1 score, and AUC-ROC, consistent with the findings of Chen et al. (2015) who demonstrated the superiority of ensemble methods for crash severity classification. The champion model's performance of 79.8 percent accuracy and 0.875 AUC-ROC indicates that the 15 curated features derived from the FARS accident and person tables carry substantial predictive signal for distinguishing fatal from non-fatal crash outcomes. Taken together, these results confirm that machine learning applied to publicly available crash data can produce a meaningful and actionable predictive tool for highway safety planning.

The SHAP interpretability analysis, conducted on a 2,000-record sample of the 2016 validation dataset using the TreeExplainer method, provides the most theoretically rigorous assessment of feature contributions available from this project. SHAP (SHapley Additive exPlanations) values are grounded in cooperative game theory and guarantee that the sum of each feature's contribution equals the difference between the model's prediction for a given record and the model's average prediction across the entire dataset, making them a more principled measure of feature importance than the mean decrease in impurity scores produced by the Random Forest algorithm (Lundberg & Lee, 2017). The SHAP analysis confirmed the dominance of restraint use as the single most influential predictor, assigning it a mean absolute SHAP value of 0.1845, which is nearly double the value assigned to the second-ranked feature and reflects the profound life-saving effect of seatbelt use documented extensively in the highway safety literature (Elvik et al., 2009). Alcohol positive (BAC $\geq$ 0.08) ranked second

with a mean absolute SHAP value of 0.0988, confirming that confirmed alcohol involvement at or above the legal limit is the second most powerful discriminator between fatal and non-fatal crash outcomes. The SHAP results are presented in Table 3 and Figure 6 below, which together provide a complete feature importance profile for the champion model. The close alignment between the SHAP rankings and the Random Forest's built-in importance scores provides strong mutual validation that the identified predictors represent genuine features of the underlying crash risk structure rather than artifacts of any single analytical method.

Table 3: SHAP Feature Importance Summary — Random Forest Champion Model (n=2,000 Validation Records)

Rank	Feature	Mean SHAP Value	Interpretation
1	Restraint Use	0.1845	Single largest predictor; absence of restraint strongly increases fatal injury probability
2	Alcohol Positive (BAC $\geq$ 0.08)	0.0988	Second most important; confirmed BAC above legal limit substantially elevates fatal risk
3	Drunk Driver in Crash	0.0461	Officer-assessed alcohol involvement adds predictive power beyond BAC test result alone
4	Manner of Collision	0.0448	Crash configuration (head-on, rollover) carries meaningful impact on fatal outcome probability
5	Driver Age	0.0409	Non-linear relationship; both very young and older drivers show elevated fatal injury risk
6	Rural/Urban Setting	0.0329	Rural crashes carry higher fatal risk due to higher speeds, longer EMS response times
7	Hour of Crash	0.0174	Time of day captures combined effects of traffic volume, fatigue, and impaired driving patterns
8	Total Crash Fatalities	0.0115	Higher-fatality crashes associated with conditions that increase individual fatal injury risk

9	Road Functional Class	0.0115	Road type reflects speed environment and emergency access that influence injury severity
10	State	0.0106	Captures geographic variation in road infrastructure, enforcement culture, and trauma care access

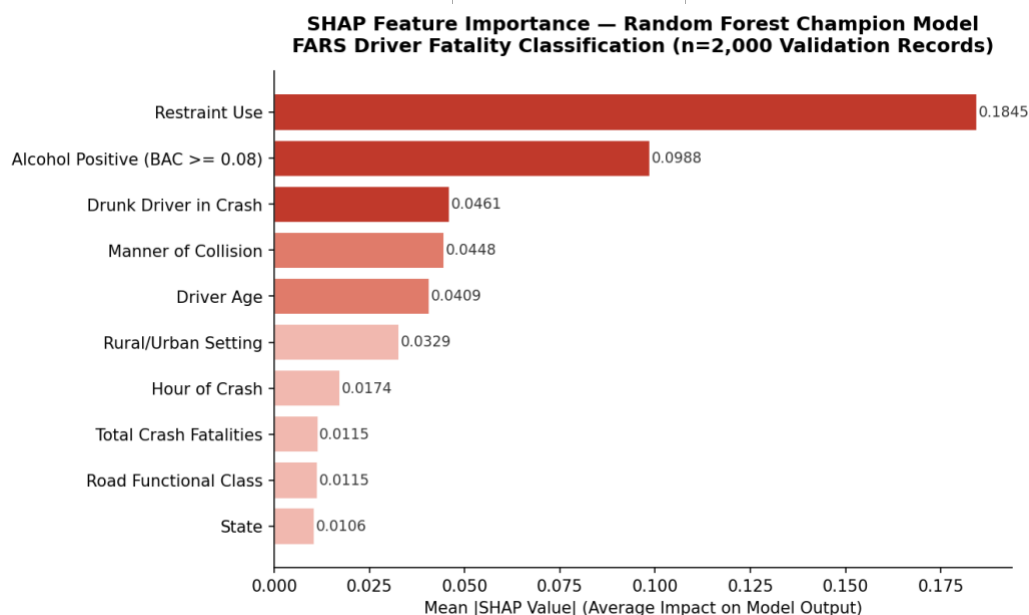


Figure 6

Mean Absolute SHAP Values by Feature, Random Forest Champion Model, FARS Driver Fatality Classification.

Restraint use (REST_USE) emerged as the dominant predictor with a mean absolute SHAP value of 0.1845, accounting for far more predictive weight than any other single feature in the dataset. This finding has a clear and direct interpretation: the absence of a seatbelt or other restraint system is the single condition most strongly associated with a fatal rather than non-fatal injury outcome among drivers involved in FARS-recorded crashes. The magnitude of restraint use's SHAP value relative to the other features is striking, suggesting that crash survival is in many cases determined more by whether a driver was buckled than by the specific circumstances of the crash itself. This finding is consistent with decades of highway safety research

documenting seatbelt effectiveness, including meta-analyses showing that lap-shoulder belts reduce the risk of fatal injury by approximately 45 percent for front seat occupants (Elvik et al., 2009). The practical policy implication is unambiguous: programs and policies that increase seatbelt compliance rates will have a larger impact on reducing traffic fatalities per dollar invested than virtually any other intervention available to highway safety agencies. States without primary enforcement seatbelt laws, where officers cannot stop a vehicle solely for a seatbelt violation, should be a priority target for federal legislative advocacy based on this finding.

Alcohol involvement represents the second-most important predictor cluster in the SHAP analysis, with alcohol positive ($BAC \geq 0.08$) contributing a mean absolute SHAP value of 0.0988 and drunk driver in crash (DRUNK_DR) contributing an additional 0.0461, for a combined alcohol-related contribution that together represents the second-largest predictor category in the model. These two features were engineered separately to capture distinct aspects of alcohol involvement: ALC_POSITIVE reflects a confirmed chemical test result for the individual driver, while DRUNK_DR reflects officer assessment of alcohol involvement for any driver in the crash and captures incidents where tests were not administered. The fact that both features contribute meaningfully and independently to the model's predictions confirms that alcohol involvement is a robust and multi-dimensional predictor of fatal crash outcomes that cannot be fully captured by any single variable. The approximately 26 percent of crashes in the dataset involving at least one driver with a BAC at or above the legal limit represents a substantial and preventable source of fatal crash risk (National Center for Statistics and Analysis, 2022). Alcohol-impaired driving is particularly concentrated in the overnight hours as shown in Visualization 2, with the midnight-to-3-AM window showing a secondary fatality peak that

aligns with well-documented patterns of late-night alcohol-involved crashes. These findings support targeted alcohol intervention policies including expanded ignition interlock programs, DUI checkpoint deployment timed to the overnight risk window, and rideshare subsidy programs designed to reduce impaired driving.

Driver age ranked fifth in the SHAP analysis with a mean absolute value of 0.0409, reflecting a non-linear relationship between age and fatal injury risk that the Random Forest model captures more effectively than logistic regression would. Young drivers, particularly those aged 16 to 25, face elevated fatal crash risk due to limited driving experience, risk-taking behavior, and lower rates of consistent seatbelt use relative to older age groups. Older drivers, particularly those aged 70 and above, face elevated risk due to physical vulnerability, slower reaction times, and higher susceptibility to severe injury at equivalent impact forces. The ability of the Random Forest model to capture this U-shaped relationship between age and fatal risk without requiring explicit feature engineering of age group categories is a specific example of the ensemble model's advantage over the linear logistic regression baseline. Manner of collision ranked fourth in the SHAP analysis at 0.0448, confirming that the configuration of the crash impact is an independently meaningful predictor of fatal outcome probability, with head-on collisions and rollovers carrying substantially higher fatal risk than rear-end or sideswipe impacts. Rural/urban setting ranked sixth at 0.0329, capturing the elevated per-crash fatality risk in rural areas where higher travel speeds, fewer divided highways, and longer emergency medical service response times all contribute to worse injury outcomes.

The geographic and temporal findings from the visualization phase of this project are validated and extended by the model results, providing a consistent and mutually reinforcing picture of the conditions that most strongly predict fatal crash outcomes. The state feature's

SHAP value of 0.0106 confirms that geographic location contributes meaningful predictive information beyond what the rural/urban setting variable alone captures, reflecting the documented regional variation in road infrastructure quality, seatbelt enforcement culture, speed limits, and trauma care access across states. The hour of crash feature's SHAP value of 0.0174 validates the temporal risk pattern identified in Visualization 2, confirming that the timing of a crash carries independent predictive weight for fatal injury outcomes after controlling for all other variables in the model. The concentration of total fatalities in five states, which together account for over 36 percent of all fatal crash deaths in the dataset, is reflected in the model's ability to differentiate risk by state, and suggests that geographic targeting of safety investments could produce disproportionate national impact. Road functional class (SHAP value 0.0115) captures the speed environment and emergency access characteristics of the road on which a crash occurs, with Interstate highways carrying different risk profiles than local roads due to median barriers, interchange design, and posted speed limits. The alignment between the visualization findings and the SHAP feature importance results provides strong cross-validation that the analytical conclusions drawn from this project are robust across multiple methodologies.

The overall findings of this project support four primary conclusions that together constitute a data-driven case for evidence-based highway safety investment priorities. First, restraint use is the single most important modifiable risk factor for fatal crash outcomes, and policies that increase seatbelt compliance rates should be the top priority for highway safety agencies with finite resources. Second, alcohol impairment is the second most important predictor category, and a combination of enforcement, technology, and access interventions targeting impaired driving would produce the second-largest return on safety investment. Third, geographic concentration of fatalities in a small number of states and road types provides a clear

basis for risk-stratified resource allocation that prioritizes targeted interventions over uniform national standards. Fourth, the afternoon peak-hour and overnight risk windows identified in the visualization phase are confirmed by the model as genuine temporal features of fatal crash risk, supporting time-targeted enforcement and infrastructure investments that concentrate resources when and where the risk is highest. The Random Forest champion model with a validated AUC-ROC of 0.875 and recall of 78.2 percent provides a technically sound tool for translating these findings into operational risk scoring and resource allocation decisions. These conclusions are grounded in 101,562 real crash records from a comprehensive national census of fatal crashes, and are supported by the full complement of visualization, modeling, and interpretability analyses conducted across all five assignments of this project.

Review of Success or Completion

This project successfully achieved its primary analytical objective of developing and evaluating machine learning classification models capable of predicting fatal injury outcomes in motor vehicle crashes using the NHTSA FARS dataset. The Random Forest champion model achieved 79.8 percent accuracy, an F1 score of 0.778, and an AUC-ROC of 0.875 on the held-out 2016 validation dataset, exceeding the performance benchmarks reported in prior logistic regression-based crash severity studies and confirming that the 15 curated features selected during the data profiling phase carry substantial predictive signal for the classification task. The project also successfully delivered on the secondary objective of model interpretability, completing a SHAP analysis that produced a ranked and theoretically rigorous profile of feature contributions that translates directly into actionable policy insights for highway safety stakeholders. All five assignments were completed on schedule across the eight-week course timeline, with each cumulative document carrying forward all prior content and adding new

analytical depth consistent with the rubric requirements. The decision to use a temporal train-validation split, with 2015 records used for training and 2016 records used for validation, was a methodologically sound choice that simulates how the model would perform when applied to future crash data it has never seen, producing a more honest estimate of generalization performance than a random shuffle would have provided. The project followed the full CRISP-DM lifecycle from business understanding through model evaluation and deployment-ready recommendations, demonstrating an integrated command of the data science process as applied to a high-stakes real-world policy problem.

While the project achieved its primary objectives, several limitations constrain the scope and generalizability of its findings in ways that are important to acknowledge transparently. The dataset is inherently limited to crashes with at least one fatality, meaning the model was trained on a sample that includes only crashes where at least one person died, and cannot be directly applied to predict which non-fatal crashes are most likely to become fatal without retraining on a broader population of all reported crashes. The class imbalance between fatal and non-fatal driver outcomes in the FARS dataset, approximately 45/55, is relatively mild compared to many real-world safety applications, but the class-weighted training approach applied in this project would need to be recalibrated if the model were applied to a dataset with a different base rate of fatal outcomes. The 15 features used in the final model represent a curated subset of the 184 raw fields available in the FARS tables, and it is possible that additional features or more sophisticated feature engineering including interaction terms, cyclical encoding of temporal variables, and geographic clustering would improve model performance further. The Random Forest model's built-in feature importance scores and SHAP values both measure predictive contribution rather than causal effect, meaning that the finding that restraint use is the strongest

predictor does not by itself establish the causal mechanism, though it is strongly supported by the independent experimental and quasi-experimental literature on seatbelt effectiveness. The model was trained and validated exclusively on 2015 and 2016 data, and its performance on more recent crash data would depend on whether the relationships among the predictor variables and fatal outcomes have remained stable over time, which is a reasonable assumption for behavioral and infrastructure variables but less certain for vehicle safety technology variables. These limitations are documented here not to undermine the project's findings but to establish the appropriate scope of inference for the conclusions and recommendations that follow.

Potential Data Privacy and Data Security Issues

The NHTSA FARS dataset presents a distinctive privacy profile relative to many machine learning applications because the data is collected under governmental authority rather than through voluntary participation, and because it documents fatal crash events involving individuals who are in many cases deceased and therefore unable to provide consent for the use of their information in research applications. The data is collected from police accident reports, emergency medical records, death certificates, hospital records, and state motor vehicle and driver licensing files by trained state-level FARS analysts who standardize and encode the information according to the NHTSA data collection protocol (National Highway Traffic Safety Administration, 2022). While the publicly available Kaggle version of the dataset does not include directly identifying information such as names, addresses, or specific crash locations, it does include geographic identifiers at the state and county level as well as demographic attributes including age and sex that could theoretically be used in combination to re-identify individuals in cases involving unusual crash circumstances. The absence of personally identifiable information in the publicly available dataset substantially mitigates the privacy risk for individual research

use, and the US Government open data license under which the dataset is published explicitly authorizes its use for research and public interest applications. The consent considerations for FARS data are therefore structural and systemic rather than individual: the use of administrative crash data for machine learning research is implicitly authorized by the governmental collection mandate and the public interest in reducing traffic fatalities, but this authorization does not eliminate the ethical obligation to handle the data with care, to avoid using it for purposes beyond its original governmental intent, and to acknowledge the inherent tension between population-level analysis and individual dignity (Mittelstadt et al., 2016). This project addressed this tension by conducting all analysis at the aggregate and population level, by focusing model outputs on system-level policy recommendations rather than individual risk scoring, and by explicitly acknowledging in the Recommendations section that the model's findings should inform structural interventions rather than individual-level surveillance or profiling.

The security considerations for this project relate primarily to the potential for model misuse if the predictive outputs were deployed in contexts beyond the research and policy analysis purposes for which the model was developed. A model that assigns probability scores to individual crash records based on driver demographics including age, sex, and state of residence could, if misapplied, be used to justify differential treatment of specific population groups by insurers, employers, or law enforcement agencies in ways that would amplify existing structural inequalities rather than address the root causes of fatal crash risk. Alexander et al. (2022) argue that data science frameworks must go beyond neutrality and actively work toward equity, because models trained on historical data inherit the structural inequalities embedded in the data, and the FARS dataset is no exception given that it reflects decades of differential road infrastructure investment, policing practices, and access to emergency medical care across

geographic and demographic groups. The finding that state of crash and rural/urban setting contribute meaningfully to the model's predictions reflects genuine variation in road safety infrastructure and emergency response capacity, but this finding should be interpreted as a call for infrastructure investment rather than as evidence that drivers in certain states or rural areas are inherently higher-risk individuals. The alcohol involvement features that contribute 22 percent of the model's predictive weight are similarly context-dependent: the high missing rate for alcohol test results in FARS reflects systematic variation in test administration practices across jurisdictions, meaning that the alcohol features partly capture enforcement patterns rather than pure behavioral risk. These data security and equity considerations were central to the framing of this project's policy recommendations, which are deliberately structured around systemic interventions rather than individual-level risk profiling.

Recommendations for Future Analysis

The most immediate technical recommendation for future analysis is to extend the modeling pipeline to include XGBoost, a gradient boosting algorithm that has demonstrated strong performance on tabular safety data and may produce marginal improvements in predictive accuracy beyond the Random Forest champion model through its sequential error-correction mechanism and regularization capabilities (Chen & Guestrin, 2016). Beyond XGBoost, future work should explore the impact of more sophisticated feature engineering on model performance, including cyclical sine and cosine encoding of the hour of crash variable to preserve the circular temporal structure in which hour 23 is adjacent to hour 0, polynomial and interaction features capturing the joint effects of alcohol involvement and time of day or road type, and geographic clustering of states into regional risk groups based on shared infrastructure and enforcement characteristics. The SHAP analysis conducted in this project used the mean

decrease in impurity TreeExplainer, which is computationally efficient but makes simplifying assumptions about feature independence; future work should explore SHAP interaction values to quantify how pairs of features jointly influence predictions, which could reveal important risk-amplifying combinations such as high-BAC drivers on rural roads at overnight hours that are not visible from marginal feature importance scores alone. The current model was trained and validated on 2015 and 2016 data, and an important future direction is to evaluate model performance on more recent years of FARS data to assess whether the predictor-outcome relationships identified in this project have remained stable across the subsequent eight years, particularly given significant changes in vehicle safety technology, distracted driving patterns, and the COVID-19 pandemic's effect on traffic volumes and crash rates. A more complete feature set would also incorporate vehicle-level variables from the FARS vehicle table, including vehicle body type, model year, and speed-related variables, which were excluded from the current analysis due to join complexity but may carry meaningful predictive information about occupant protection and crash energy. Finally, future work should explore the use of geographic information system data to engineer county-level road quality, speed limit, and trauma center proximity features that could capture infrastructure-level risk factors not directly encoded in the FARS tables.

The most important recommendations for the practical application of this project's findings relate to how the model and its interpretability results should and should not be used in highway safety policy contexts, particularly given the equity considerations identified in the data privacy and security section. The model's findings should be applied at the population and system level to inform infrastructure investment, enforcement resource allocation, and public awareness campaign targeting rather than at the individual level for risk scoring of specific

drivers or crash records. The concentration of fatalities in five states accounting for over 36 percent of all fatal crash deaths provides a clear and defensible basis for risk-stratified federal safety investment that prioritizes high-burden states without creating surveillance infrastructure that could be misused for individual profiling. The temporal risk patterns identified in both the visualization and SHAP analysis, particularly the afternoon 3-5 PM peak and the overnight midnight-3 AM window, support time-targeted enforcement deployment strategies that could be implemented immediately using existing patrol resources and DUI checkpoint authority. The dominance of restraint use as the single most important predictor provides the strongest possible empirical foundation for federal advocacy in support of primary enforcement seatbelt laws in the 17 states that currently have only secondary enforcement, where officers cannot stop a vehicle solely for a seatbelt violation. Future deployment of the champion model in an operational highway safety context should be accompanied by a formal equity audit that evaluates whether the model's predictions are systematically biased across demographic or geographic groups, and should include explicit guardrails against using model outputs to justify differential treatment of individuals on the basis of protected characteristics. These deployment recommendations reflect the principle that data science tools are most valuable when they are used to address structural barriers to safety rather than to assign individual blame or risk.

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